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1.

```
with open ('file.txt', 'r') as f:
    students = f.readlines()

students = [x.strip() for x in students]
print(students)
```

```
PS E:\Pranav\IMCC\Assignments\Python\1> py 1.py
['Pranav', 'Madhav', 'Vinayak', 'Ram', 'Aayush', 'Pranav', 'Aabaa', 'Ram']
PS E:\Pranav\IMCC\Assignments\Python\1>
```

2.

```
with open ('file.txt', 'r') as f:
    students = f.readlines()

students = [x.strip() for x in students]

for i in students:
    if len(i) > 5:
        print(i)
```

```
PS E:\Pranav\IMCC\Assignments\Python\1> py 2.py
Pranav
Madhav
Vinayak
Aayush
Pranav
PS E:\Pranav\IMCC\Assignments\Python\1>
```

,

3.

```
with open('file.txt', 'r') as f:
    students = f.readlines()

vow = {'a', 'e', 'i', 'o', 'u'}
count = 0
for i in students:
    if i[0].lower() in vow:
        count += 1

print(f"{count} students name starts with vowels")
```

```
PS E:\Pranav\IMCC\Assignments\Python\1> py 3.py
2 students name starts with vowels
PS E:\Pranav\IMCC\Assignments\Python\1>
```

4.

```
with open('file.txt', 'r') as f:
    students = f.readlines()

students = [student.strip() for student in students]
students = filter(lambda x: len(x)> 5, students)

with open('students.txt', 'w') as f2:
    for i in students:
        f2.write(i + "\n")
```

5.

```
with open('file.txt', 'r') as f:
    students = f.readlines()

students = [student.strip() for student in students]

for i in range(len(students)):
    for j in range(i+1, len(students)):
        if students[i] > students[j]:
            students[i], students[j] = students[j], students[i]
```

```
for i in students:
    print(i)
```

```
PS E:\Pranav\IMCC\Assignments\Python\1> py 5.py
Aabaa
Aayush
Madhav
Pranav
Pranav
Ram
Ram
Vinayak
PS E:\Pranav\IMCC\Assignments\Python\1>
```

6.

```
with open('file.txt', 'r') as file:
    students = file.readlines()

students = [student.strip() for student in students]
studentsWithLetterA = list(filter(lambda x: 'a' in x.lower(), students))

for i in studentsWithLetterA:
    print(i)
```

```
PS E:\Pranav\IMCC\Assignments\Python\1> py 6.py
Pranav
Madhav
Vinayak
Ram
Aayush
Pranav
Aabaa
Ram
PS E:\Pranav\IMCC\Assignments\Python\1>
```

7.

```python
with open("file.txt", "r") as f:
    students = f.readlines()

students = [student.strip() for student in students]

d = dict()
for i in students:
    if i in d:
        d[i] += 1
    else:
        d[i] = 1

for student, count in d.items():
    print(f"{student}: {count}")
```

```
PS E:\Pranav\IMCC\Assignments\Python\1> py 7.py
Pranav: 2
Madhav: 1
Vinayak: 1
Ram: 2
Aayush: 1
Aabaa: 1
PS E:\Pranav\IMCC\Assignments\Python\1>
```

8.

```python
def isPalindrome(s):
    s = s.lower()
    return s == s[::-1]

with open("file.txt", "r") as f:
    students = f.readlines()

students = [student.strip() for student in students]

for i in students:
    print(f"Length of {i}: {len(i)}")
    print(f"Name {i} is {'a' if isPalindrome(i) else 'not a'} palindrome")
    print(f"First Letter {i[0]} and Last Letter {i[-1]}")
    print()
```

```
PS E:\Pranav\IMCC\Assignments\Python\1> py 8.py
Length of Pranav: 6
Name Pranav is not a palindrome
First Letter P and Last Letter v

Length of Madhav: 6
Name Madhav is not a palindrome
First Letter M and Last Letter v

Length of Vinayak: 7
Name Vinayak is not a palindrome
First Letter V and Last Letter k

Length of Ram: 3
Name Ram is not a palindrome
First Letter R and Last Letter m

Length of Aayush: 6
Name Aayush is not a palindrome
First Letter A and Last Letter h

Length of Pranav: 6
Name Pranav is not a palindrome
First Letter P and Last Letter v

Length of Aabaa: 5
Name Aabaa is a palindrome
First Letter A and Last Letter a

Length of Ram: 3
Name Ram is not a palindrome
First Letter R and Last Letter m

PS E:\Pranav\IMCC\Assignments\Python\1>
```

9.

```python
with open("file.txt", "r") as f:
    students = f.readlines()

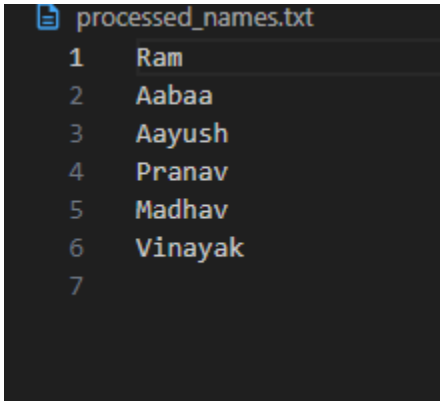
students = [student.strip() for student in students]

distinct_students = list(set(students))

for i in range(len(distinct_students)):
    for j in range(i + 1, len(distinct_students)):
        if len(distinct_students[i]) > len(distinct_students[j]):
```

```
            distinct_students[i], distinct_students[j] =
distinct_students[j], distinct_students[i]

with open("processed_names.txt", "w") as f:
    for student in distinct_students:
        f.write(student + "\n")
```



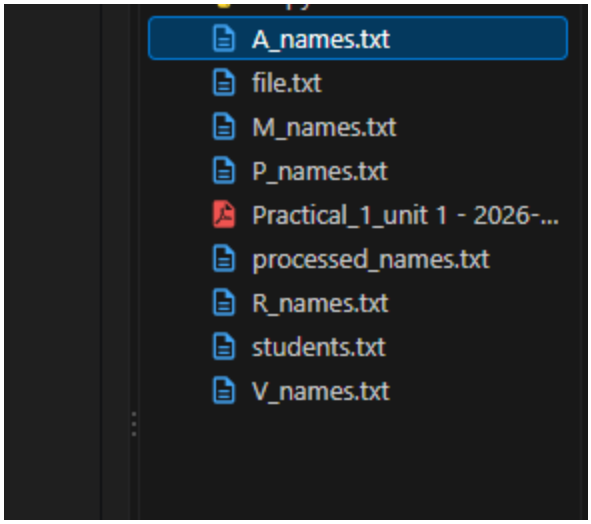
10.

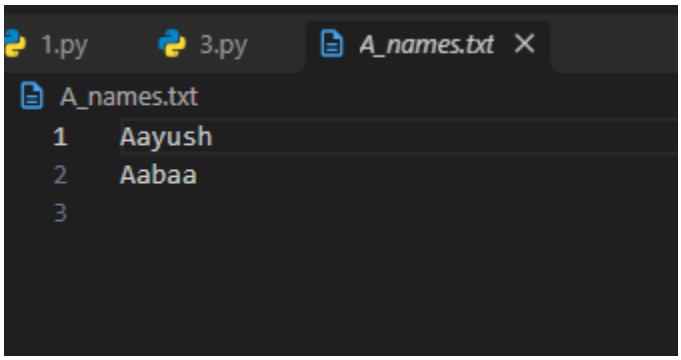
```
with open("file.txt", "r") as f:
    students = f.readlines()

students = [student.strip() for student in students]

d = dict()
for i in students:
    if i[0] in d:
        d[i[0]].append(i)
    else:
        d[i[0]] = [i]

for i, names in d.items():
    with open(f"{i.upper()}_names.txt", "w") as f:
        for name in names:
            f.write(name + "\n")
```


A_names.txt
file.txt
M_names.txt
P_names.txt
Practical_1_unit 1 - 2026-...
processed_names.txt
R_names.txt
students.txt
V_names.txt


1.py
3.py
A_names.txt
A_names.txt
1 Aayush
2 Aabaa
3